

Estimating total sardine recruitment and proportions by stock and coast: further work towards selecting prior distributions

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Alternative beta prior distributions are considered for estimating the proportional split of total sardine recruitment by stock and coast. This follows a recent change in the sardine assessment to estimate total annual sardine recruitment and then proportion that by stock and coast. A symmetrical, uninformative prior distribution is preferred for the proportion of WTS recruitment which recruits to the west coast. Either a symmetrical, uninformative, or an informative beta prior distribution, centred around ~ 0.29 , could be used for the proportion of total recruitment which is WTS. The first set of projections will therefore consider both alternatives.

Keywords: beta distribution, parameter estimation, population dynamics, recruitment, sardine, South Africa

Introduction

In modelling the South African sardine population, de Moor (2025a) estimated (i) annual recruitment of cold temperate sardine (CTS), (ii) annual recruitment of warm temperate sardine (WTS) and (iii) annual proportion of WTS which recruit to the west coast. The international review panel suggested that an alternative be tried, in which the total annual recruitment of both stocks is estimated, with proposals for how the proportions by stock/coast could be estimated with potentially more informative priors than those used by de Moor (2025a). Following the results of de Moor (2026b), the Small Pelagic Scientific Working Group chose to estimate (i) annual total sardine recruitment, R_y , (ii) the proportion of total sardine recruitment which is CTS, q_y , and (iii) the proportion of WTS sardine recruitment which recruits to the west coast, p_y . This document considers further alternative prior distributions for q_y and p_y .

Methods

The models used in this document assume lognormal likelihoods for survey estimates of abundance and multinomial likelihoods for survey and commercial proportion-at-length data with $N_{propl}^{sur} = 3$ and $N_{propl}^{com} = 0.75$, i.e. an extension of Sc of de Moor (2026a). A range of symmetrical and a-symmetrical beta prior distributions are considered for p_y and q_y (Figure 1). The means of the a-symmetrical prior distributions were selected from the realised means of p_y and q_y assuming $p_y, q_y \sim \text{Beta}(1.1, 1.1)$ (de Moor 2026a).

Results and Discussion

Table 1 gives a summary of key statistics that could be used to select among the alternative prior distributions; these statistics mostly focus on the period after the last pulse and exclude the final estimated year of recruitment, i.e. 2005-2023:

- i) The negative log likelihood is the fit to the data - smaller numbers correspond to better fits.
- ii) The median CV on individual estimates of recruitment - should be at least less than 1 (IWS 2025 panel pers. comm) except in initial and final years ("bowl plots").
- iii) The CV of the timeseries of CTS / WTS recruitment should not be too much higher than that for total recruitment – a check to see if the difference is ≤ 0.2 is included.

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- iv) The variance of the timeseries of CTS+WTS recruitment should be close to that of the total recruitment if the CTS and WTS timeseries are independent. A multiplicative comparison is given, so numbers close to 1 would be considered better.
- v) The correlation coefficient between the timeseries of CTS and WTS recruitment, adding further context to (iv).
- vi) The estimated multiplicative bias for the hydro-acoustic November and recruit surveys; the prior for $\ln(k_{ac})$ has an effective mean of 0.80 for k_{ac} , previous posterior estimates for the sardine models have been around ~ 0.70 for k_{ac} and ~ 0.3 for $k_{c,r}$.

More detailed results are given in Tables A1 and A2.

The fits to the data (Table 1, A1) indicate that a-symmetrical prior distributions with a mean of 0.67 for p_y are resulting in convergence difficulties. Preference should therefore be given to symmetrical prior distributions for p_y . As expected, the model is able to fit the data better for less informative prior distributions on q_y .

The timeseries of estimated recruitment is given in Figure 2. The different absolute scale in these timeseries primarily reflects a difference in the estimated multiplicative bias parameters. To assist further with choosing between alternatives, the models with a mean for the beta prior on q_y of 0.29 were run with a fixed $k_{ac} = 0.8$, corresponding to the effective mean of the prior distribution for k_{ac} . This gave the following -lnL values:

SD	$k_{ac} = 0.8$	k_{ac} estimated
0.10	1487	1485 ($k_{ac} = 0.99$)
0.15	1478	1483 ($k_{ac} = 0.88$)
0.18	1475	1475 ($k_{ac} = 0.78$)
0.20	1473	1473 ($k_{ac} = 0.73$)
0.25	1527	

These indicate that an SD of 0.18 is likely the best trade-off between fitting the data and estimated k_{ac} is not too far from the mean of the prior distribution. An SD of 0.20 provides an improved fit to the data with an estimate of k_{ac} that is still within the range of that previously estimated for South African sardine.

The estimated proportions q_y and p_y are shown in Figures 3 and 4, with the a-symmetrical prior distributions resulting in a substantial reduction in q_y , particularly prior to 2000. Note that the realised $\bar{q}_y$ is less than 0.29 for all cases considered (Table A1). The estimated proportions from the symmetrical prior distributions follow a similar pattern to that previously estimated (de Moor 2025b), but with less extreme values. The greater inter-annual variability associated with the less-informative prior distributions corresponds with the improved fit to the data (Table 1, A1). However, these 'extremes' may not be well informed by the data. Figure 5 shows the corresponding estimates of the proportion of 0-year old WTS which move from the west to the south coast.

Figure 6 shows the hessian-based CVs for annual estimates of recruitment ('bowl plots'). The medians are given in Table 1. High CVs are expected at the start and end of the timeseries given there is less data to inform these estimates of recruitment. However, there are also high CVs (an indication that the parameters are not precisely estimated), particularly

for CTS, for the more uninformative prior distributions. These CVs are more reasonable for the symmetrical priors $q_y \sim \text{Beta}(x, x)$, with $x > 1.2$ with improvements for larger values of x , and for the a-symmetrical priors with SDs ≤ 0.18 .

The final set of comparisons to help choose between alternative concerns the CV and variance of the timeseries of estimates of recruitment. Note this is different from the CV of the individual estimates given in Figure 6, but instead reflects the CV and variance of the timeseries plotted in Figure 2. Ideally, the CV for the recruitment of CTS and WTS should not be too much bigger than that for the total population. This is not satisfied for the more uninformative prior distributions. If the time series for the two stocks are independent, one could expect the variance of the timeseries of total recruitment to be close to (not much less than) that of the sum of the two stocks. However, using this statistic to select among alternatives is not straightforward as the correlation between the stocks increases as the prior distributions become more informative (Table 1, Figure 7).

As a means to check the robustness of using a mean of 0.29 for the a-symmetrical beta prior on q_y , alternatives were run with higher and lower means:

Mean of prior on q_y		0.24	0.29	0.34
		-lnL		
Standard deviation	0.15	1477	1483	1480
	0.18	1474	1475	1476 (nonPDH)
		Realised $\bar{q}_y$		
Standard deviation	0.15	0.14	0.21	0.27
	0.18	0.08	0.17	

These do not indicate a robustness to the choice of the mean, with realised $\bar{q}_y$ still being reduced from the prior's mean.

Conclusion

Taking all the above into account, the following prior distributions are suggested for the sardine model:

- 1) An a-symmetrical beta prior distribution on q_y with a mean of 0.29 and a standard deviation of 0.18, with $p_y \sim \text{Beta}(1.1, 1.1)$.
- 2) The symmetrical prior distributions $q_y, p_y \sim \text{Beta}(1.35, 1.35)$.
- 3) An a-symmetrical beta prior distribution on q_y with a mean of 0.29 and a standard deviation of 0.15, with $p_y \sim \text{Beta}(1.1, 1.1)$.

Following discussions in the Technical Task Group, options 1) and 3) will be attempted with $p_y \sim \text{Beta}(1.35, 1.35)$ to reduce differences between the alternatives.

The full set of results for these alternatives are shown in Appendix B. These are mostly in line with what has previously been estimated. However, the south coast growth curves for a beta prior on q_y with a mean of 0.29 and a standard deviation of 0.15, and for a prior of $q_y \sim \text{Beta}(1.35, 1.35)$ reflects a later peak spawning than expected *a priori* (Figure B7).

References

de Moor CL. 2025a. Revisions to the updated assessment model for the revised sardine stock structure hypothesis. DFFE: Branch Fisheries Document FISHERIES/2025/OCT/SWG-PEL/54rev.

- de Moor CL. 2025b. Results from the updated stock assessment model of the revised South African sardine stock structure hypothesis. International Stock Assessment Workshop, Cape Town, 1-5 December 2025, Document IWS/2025/Sardine/BG4.
- de Moor CL. 2026a. Changes to the likelihood of the sardine assessment. DFFE: Branch Fisheries Document FISHERIES/2026/JAN/SWG-PEL/01.
- de Moor CL. 2026b. Estimating total sardine recruitment and proportions by stock and coast. DFFE: Branch Fisheries Document FISHERIES/2026/FEB/SWG-PEL/03.

Table 1. Summary table with some comparisons to assist with selecting among alternative prior distributions for p_y and q_y : the negative log likelihood, the median of the hessian-based CVs of the estimates of recruitment from 1985 to 2022 and 2005 to 2023 (cf Figure 6), a check if the CV of the time series of CTS/WTS recruitment is ≤ 0.2 above the CV of the time series of total recruitment, the multiplicative difference between the variance of the time series of total recruitment and the sum of the variances of the time series of WTS and CTS recruitment, and the correlation between the time series of CTS and WTS estimates of recruitments (and log recruitments) from 2005 to 2023. The multiplicative bias associated with the November and recruit surveys are also given; the prior on $\ln(k_{ac})$ has an effective mean of 0.80 for k_{ac} .

Prior on p_y	Prior on q_y	Fit to the data (-lnL)	Median CV on estimates of recruitment (85-22)			Median CV on estimates of recruitment (05-23)			Is CV(CTS) $\leq$ CV(Tot)+0.2 ¹	Is CV(WTS) $\leq$ CV(Tot)+0.2 ¹	Var(Tot) / (Var(WTS)+Var(CTS)) ¹	Corelation coefficient	k_{ac}	$k_{w,r}$	$k_{s,r}$
			Tot	CTS	WTS	Tot	CTS	WTS							
$Beta(1.01,1.01)$	$Beta(1.01,1.01)$	1471	0.40	1.60	0.58	0.38	2.38	0.55	No	Yes	1.03	0.04 (-0.05)	0.73	0.29	0.28
$Beta(1.1,1.1)$	$Beta(1.1,1.1)$	1473	0.40	1.23	0.60	0.38	1.58	0.59	No	Yes	1.11	0.19 (0.21)	0.75	0.30	0.30
$Beta(1.2,1.2)$	$Beta(1.2,1.2)$	1475	0.40	1.04	0.59	0.38	1.18	0.58	No	Yes	1.14	0.24 (0.31)	0.78	0.32	0.31
$Beta(1.3,1.3)$	$Beta(1.3,1.3)$	1477	0.40	0.91	0.59	0.38	1.03	0.57	Yes	Yes	1.15	0.27 (0.37)	0.81	0.34	0.31
$Beta(1.35,1.35)$	$Beta(1.35,1.35)$	1478	0.40	0.85	0.60	0.39	0.98	0.57	Yes	Yes	1.16	0.29 (0.39)	0.82	0.35	0.30
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.1$	1485	0.36	0.49	0.40	0.34	0.48	0.38	Yes	Yes	1.50	0.96 (0.97)	1.00	0.47	0.46
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.15$	1482	0.38	0.72	0.46	0.37	0.71	0.45	Yes	Yes	1.34	0.78 (0.82)	0.88	0.36	0.35
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.18$	1475	0.38	1.10	0.47	0.36	1.06	0.45	Yes	Yes	1.27	0.67 (0.69)	0.78	0.33	0.32
$Beta(1.1,1.1)$	$\mu=0.29, SD= 0.2$	1473	0.38	1.87	0.47	0.36	1.92	0.45	No	Yes	1.21	0.52 (0.54)	0.73	0.29	0.29
$\mu=0.67, SD=0.1$	$Beta(1.35,1.35)$	1515 ²											0.79	0.33	0.18
$\mu=0.67, SD=0.15$	$Beta(1.35,1.35)$	1499 ²											0.76	0.30	0.18
$\mu=0.67, SD=0.2$	$Beta(1.35,1.35)$	1488 ²											0.74	0.28	0.25
$\mu=0.67, SD=0.1$	$\mu=0.29, SD=0.18$	1525	0.36	1.07	0.49	0.36	1.11	0.45	No	Yes	1.16	0.35 (0.36)	0.83	0.37	0.15
$\mu=0.67, SD=0.15$	$\mu=0.29, SD=0.18$	1492 ²											0.79	0.33	0.21
$\mu=0.67, SD=0.2$	$\mu=0.29, SD=0.18$	1481 ²											0.81	0.31	0.31

¹ CV of timeseries of recruitments from 2005-2023.

² Non positive definite Hessian and thus statistics are not provided as they may not be reliable.

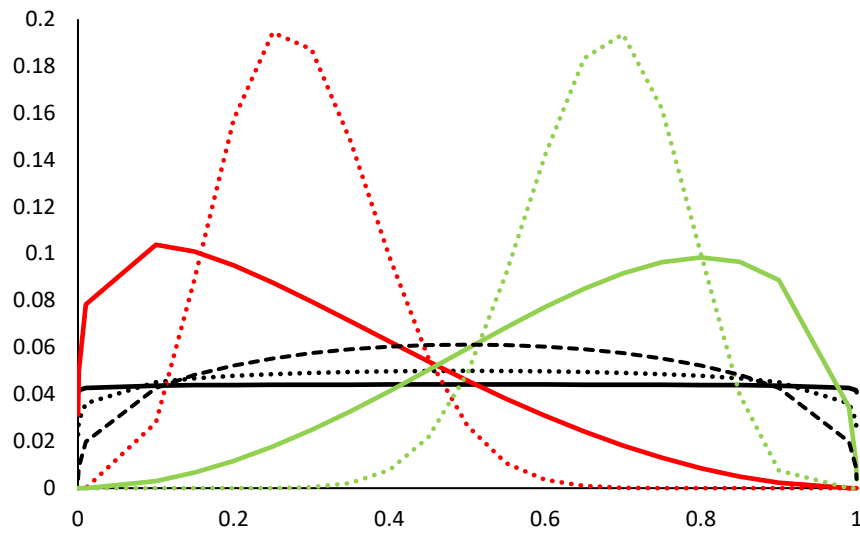


Figure 1. The beta prior distributions for (i) symmetrical distributions (black) with parameters 1.01 (dotted), 1.1 (solid) and 1.35 (dashed), (ii) distributions with a mean of 0.29 (red) and standard deviation of 0.1 (dotted) and 0.2 (solid), and (iii) distributions with a mean of 0.67 (green) and standard deviation of 0.1 (dotted) and 0.2 (solid).

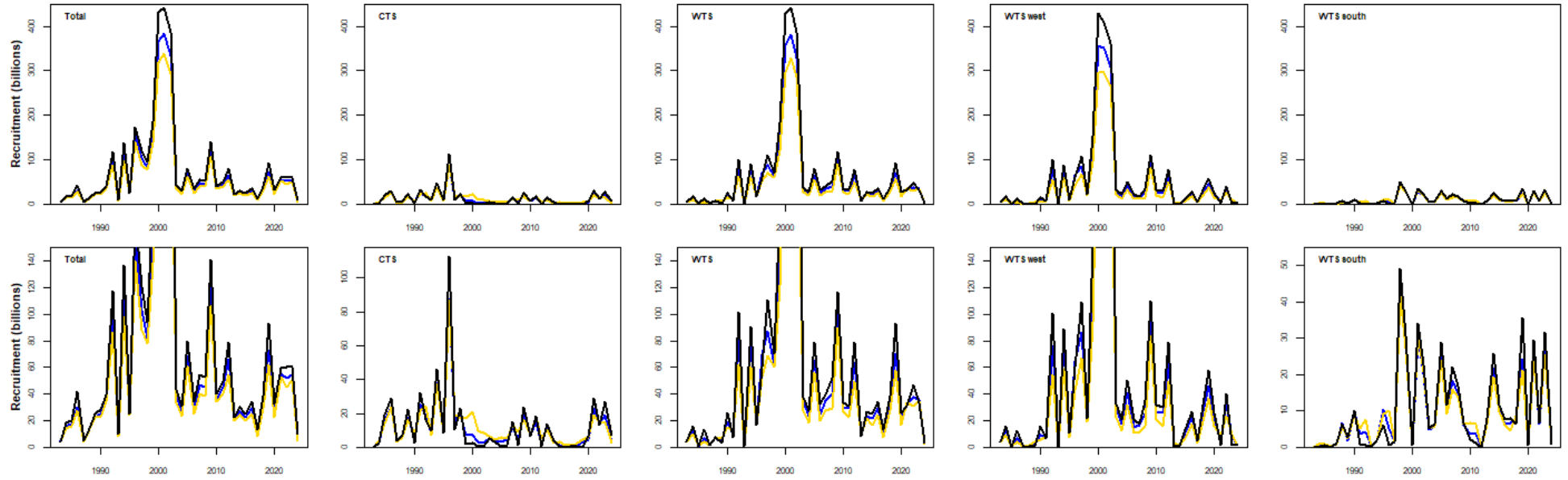


Figure 2a. The estimated total recruitment, CTS recruitment, WTS recruitment, and WTS recruitment to the west and south coasts for $p_y, q_y \sim \text{Beta}(1.01, 1.01)$ (black), $p_y, q_y \sim \text{Beta}(1.1, 1.1)$ (blue) and $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (gold). The lower row is a repeat of the upper row, but with different y-axis ranges.

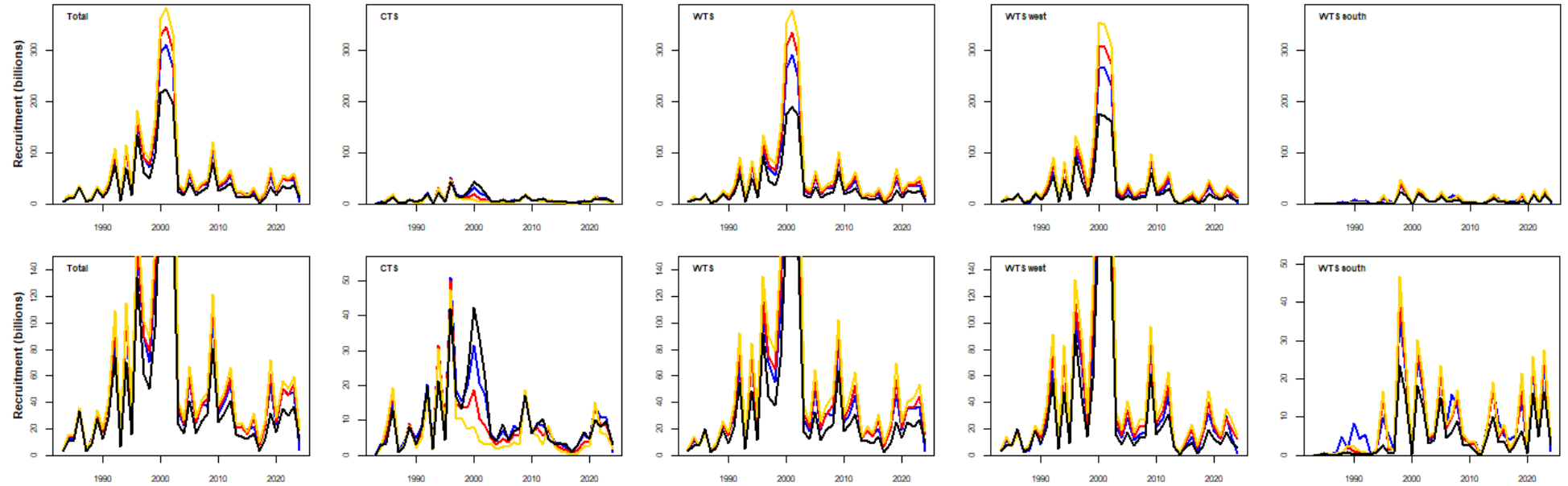


Figure 2b. The estimated total recruitment, CTS recruitment, WTS recruitment, and WTS recruitment to the west and south coasts assuming $p_y \sim \text{Beta}(1.1, 1.1)$ and a Beta distribution for q_y with mean 0.29 and standard deviation of 0.10 (black), 0.15 (blue), 0.18 (red) or 0.20 (gold). The lower row is a repeat of the upper row, but with different y-axis ranges.

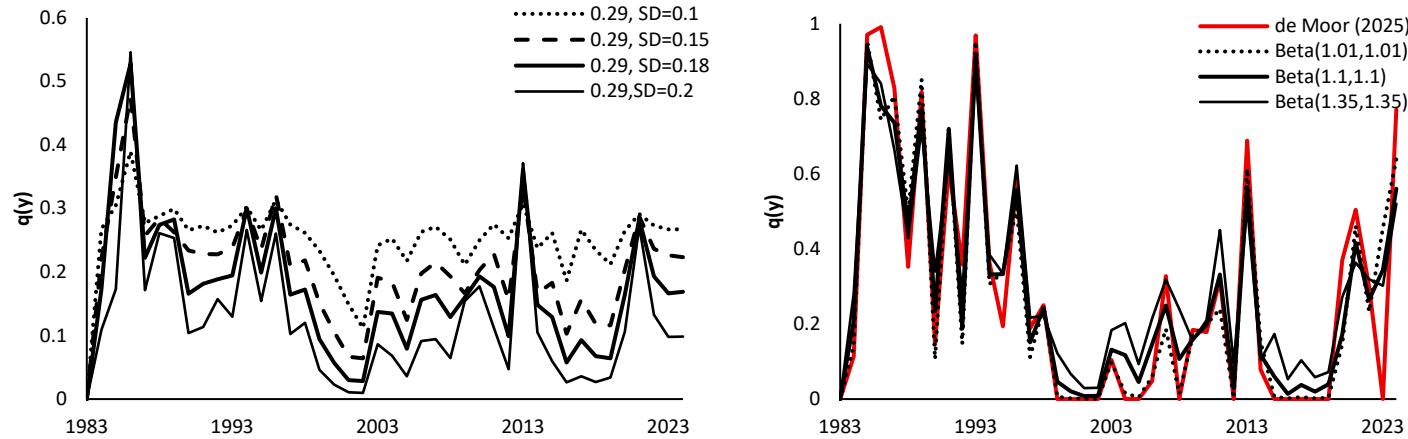


Figure 3. The estimated annual proportion of total recruitment which is CTS, q_y , using a beta prior distribution with a mean of 0.29 and a SD of 0.10, 0.15, 0.18 and 0.20 (left panel), or $q_y \sim \text{Beta}(1.01, 1.01)$, $q_y \sim \text{Beta}(1.1, 1.1)$ or $q_y \sim \text{Beta}(1.35, 1.35)$ (right panel). The CTS recruitment as a proportion of CTS+WTS recruitment, as estimated by de Moor (2025b) is also given in the right panel.

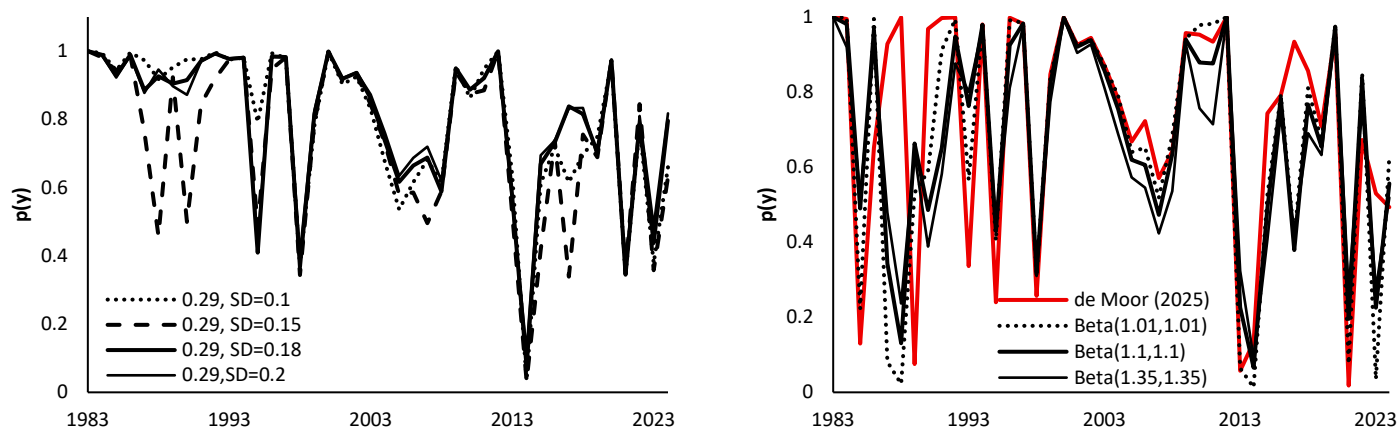


Figure 4. The estimated annual proportion of WTS recruitment which recruits to the west coast, p_y , using a beta prior distribution for q_y with a mean of 0.29 and a SD of 0.10, 0.15, 0.18 and 0.20 (left panel), or $q_y \sim \text{Beta}(1.01, 1.01)$, $q_y \sim \text{Beta}(1.1, 1.1)$ or $q_y \sim \text{Beta}(1.35, 1.35)$ (right panel). The proportion estimated by de Moor (2025b) is also given in the right panel.

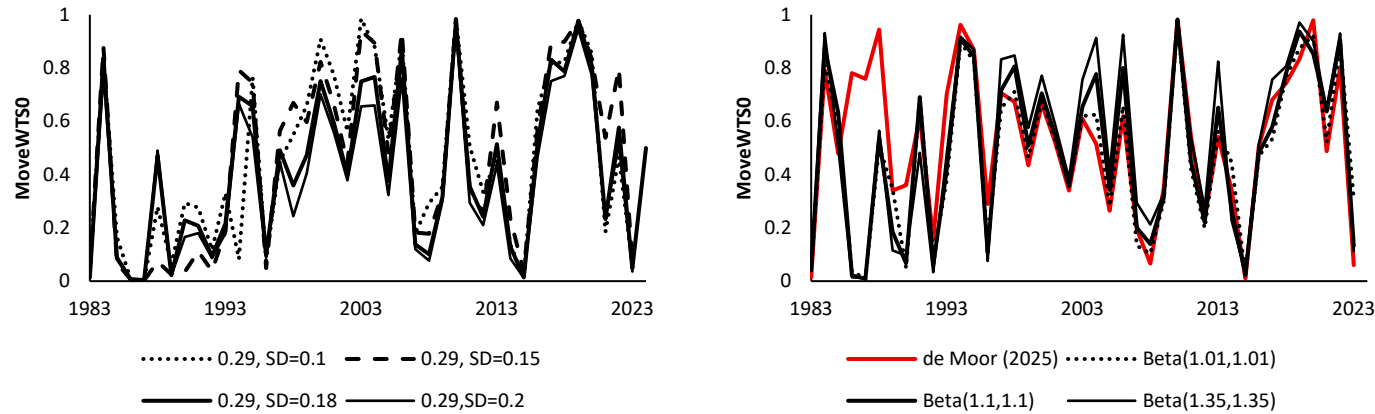


Figure 5. The estimated annual proportion of WTS 0-year olds distributed off the west coast that move to the south coast, for a beta prior distribution for q_y with a mean of 0.29 and a SD of 0.10, 0.15, 0.18 and 0.20 (left panel), or $q_y \sim \text{Beta}(1.01, 1.01)$, $q_y \sim \text{Beta}(1.1, 1.1)$ or $q_y \sim \text{Beta}(1.35, 1.35)$ (right panel). The proportion estimated by de Moor (2025b) is also given in the right panel.

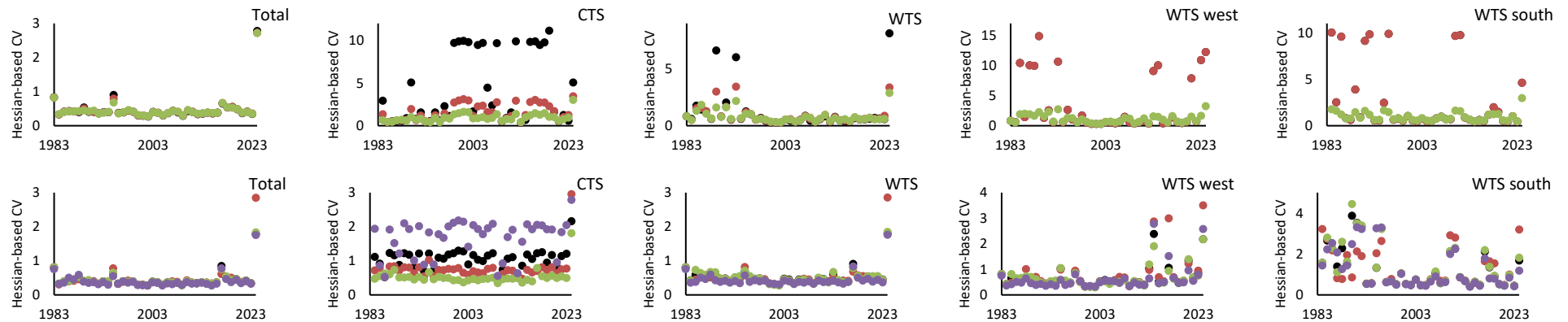


Figure 6. The Hessian-based CVs for the estimated annual (i) Total recruitment, (ii) CTS recruitment, (iii) WTS recruitment, (iv) WTS recruitment to the west coast and (v) WTS recruitment to the south coast. The top row corresponds to $p_y, q_y \sim \text{Beta}(1.01, 1.01)$ (black), $p_y, q_y \sim \text{Beta}(1.1, 1.1)$ (brown) and $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (green). The second row corresponds to a Beta distribution for q_y with mean 0.29 and standard deviation 0.20 (purple), 0.18 (black), 0.15 (brown), or 0.10 (green), with $p_y \sim \text{Beta}(1.1, 1.1)$.

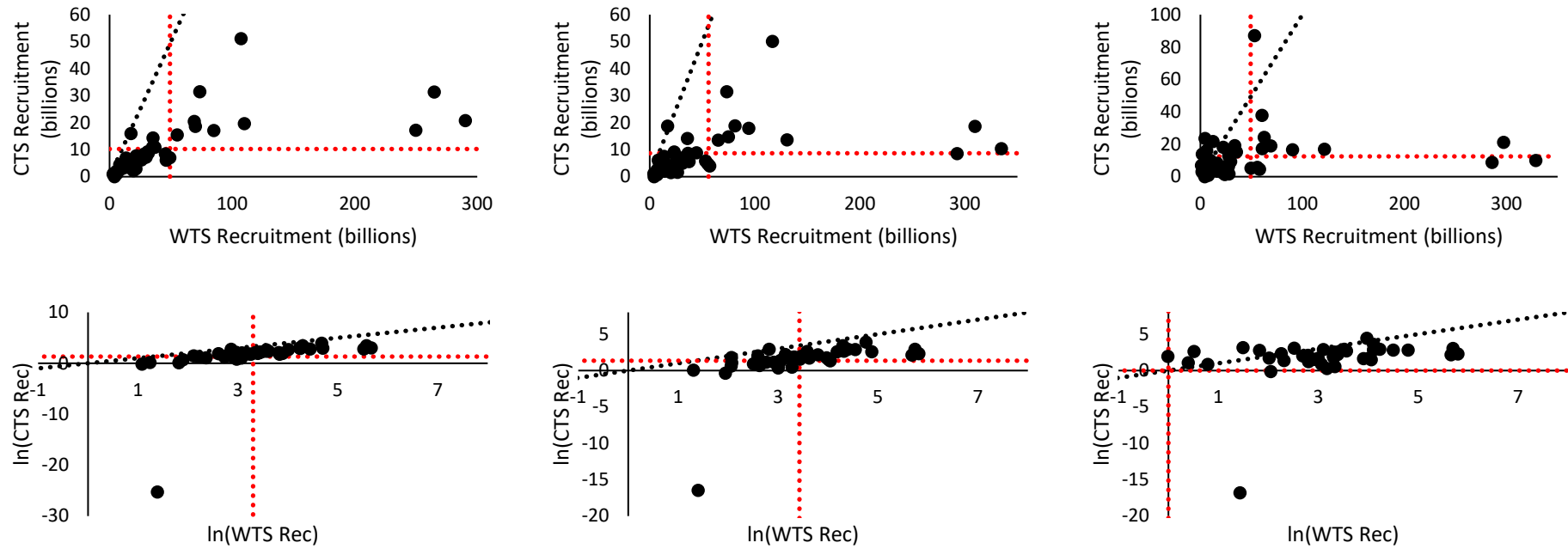


Figure 7. Annual CTS recruitment plotted against WTS recruitment, assuming a beta prior distribution for q_y with a mean of 0.29 and a SD of 0.15 (left) or 0.18 (middle) or $q_y \sim \text{Beta}(1.35, 1.35)$ (right). The red dotted lines indicate the mean of the timeseries. The black dotted line has a slope of 1. The lower row is a repeat of the top ones but in log-space.

Appendix A: Additional Results

Table A1. The contributions to the objective function from likelihood and prior components at the joint posterior mode for the models considered in this document, and S_{noSR} from de Moor (2025b), however different likelihood distributions were used in this model making direct comparisons from this table difficult.

Prior on p_y	Prior on q_y	Realised $\bar{p}_y$	Realised $\bar{q}_y$	Objective function	log likelihood						log priors						
					$-\ln L$	$-\ln L^{\text{Nov}}$	$-\ln L^{\text{rec}}$	$-\ln L^{\text{sur propl}}$	$-\ln L^{\text{com propl}}$	$-\ln L^{\text{prev}}$	$\ln(k_{ac}^s)$	$move_{y,1}$	$\bar{l}_{1,y}$	ϑ_a	S_{50}	η_{cy}^t	p_y, q_y
de Moor (2025b)		0.72	0.28	-837.0	-901.2	41.5	30.9	-495.0	-478.6	-	-0.54	4.05	56.05	0.80	2.35	0.22	1.25
$Beta(1.01,1.01)$	$Beta(1.01,1.01)$	0.67	0.27	1530.4	1471.4	40.6	28.7	791.6	610.4	-	-0.66	4.08	52.07	0.69	0.20	0.22	2.47
$Beta(1.1,1.1)$	$Beta(1.1,1.1)$	0.68	0.28	1549.2	1473.3	40.9	29.1	792.4	611.0	-	-0.70	4.26	52.04	0.68	0.18	0.22	19.27
$Beta(1.2,1.2)$	$Beta(1.2,1.2)$	0.67	0.29	1567.7	1475.2	41.1	29.4	793.2	611.5	-	-0.73	4.40	52.03	0.68	0.16	0.22	35.80
$Beta(1.3,1.3)$	$Beta(1.3,1.3)$	0.66	0.31	1585.2	1477.1	41.3	29.8	794.0	611.9	-	-0.73	4.50	52.01	0.69	0.15	0.23	51.28
$Beta(1.35,1.35)$	$Beta(1.35,1.35)$	0.66	0.31	1593.7	1478.1	41.4	30.1	794.4	612.1	-	-0.73	4.54	52.01	0.69	0.15	0.23	58.69
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.1$	0.79	0.25	1979.7	1485.3	45.2	32.5	791.9	615.7	-	0.13	4.69	52.85	0.89	1.31	0.34	434.14
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.15$	0.73	0.21	1689.6	1482.8	43.1	29.7	796.3	613.8	-	-0.48	4.95	52.33	0.75	0.14	0.23	148.88
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.18$	0.80	0.17	1612.4	1474.8	42.2	31.4	788.8	612.4	-	-0.68	4.56	53.02	0.78	2.02	0.19	77.72
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.2$	0.81	0.13	1576.1	1472.8	42.0	31.5	787.7	611.5	-	-0.66	4.52	52.98	0.77	2.00	0.18	43.53
$\mu=0.67, SD=0.1$	$Beta(1.35,1.35)$	0.70	0.21	2088.1	1514.9	41.0	50.3	801.3	622.2	-	-0.79	4.05	53.59	0.59	0.04	0.25	515.49
$\mu=0.67, SD=0.15$	$Beta(1.35,1.35)$	0.75	0.18	1758.1*	1499.2	40.8	39.6	799.4	619.5	-	-0.81	4.13	53.01	0.59	0.11	0.32	201.59
$\mu=0.67, SD=0.2$	$Beta(1.35,1.35)$	0.81	0.16	1632.4	1488.0	41.0	32.7	798.1	616.2	-	-0.77	4.17	52.78	0.60	0.09	0.29	87.25
$\mu=0.67, SD=0.1$	$\mu=0.29, SD=0.18$	0.69	0.30	2159.2	1524.5	45.4	66.6	799.1	613.5	-	-0.72	4.87	52.77	0.77	1.10	0.17	575.74
$\mu=0.67, SD=0.15$	$\mu=0.29, SD=0.18$	0.71	0.29	1811.9*	1492.3	43.3	42.0	794.5	612.5	-	-0.79	4.93	52.81	0.76	1.28	0.16	260.46
$\mu=0.67, SD=0.2$	$\mu=0.29, SD=0.18$	0.75	0.26	1670.4*	1481.4	42.0	32.9	793.8	612.6	-	-0.64	4.84	52.79	0.70	0.60	0.18	130.56

* Non positive definite hessian after a single run

Table A2. The mean, median, CV and variance of the time series of recruitment from 2005 to 2023 and from 1983 to 2024, separated into total recruitment, CTS recruitment, WTS recruitment and WTS recruitment to the west and south coasts.

		2005 to 2023 ³					1983 to 2024				
Prior on p_y	Prior on q_y	Total	CTS	WTS west	WTSs outh	WTS	Total	CTS	WTS west	WTSs outh	WTS
		CV									
$Beta(1.01,1.01)$	$Beta(1.01,1.01)$	0.55	1.21	0.90	0.83	0.61	1.29	1.43	1.73	1.17	1.51
$Beta(1.1,1.1)$	$Beta(1.1,1.1)$	0.55	0.93	0.90	0.76	0.60	1.29	1.29	1.77	1.09	1.53
$Beta(1.2,1.2)$	$Beta(1.2,1.2)$	0.55	0.80	0.93	0.73	0.61	1.29	1.21	1.81	1.05	1.54
$Beta(1.3,1.3)$	$Beta(1.3,1.3)$	0.55	0.74	0.95	0.71	0.62	1.29	1.16	1.83	1.03	1.55
$Beta(1.35,1.35)$	$Beta(1.35,1.35)$	0.55	0.71	0.96	0.71	0.62	1.28	1.14	1.84	1.03	1.55
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.1$	0.62	0.56	0.91	0.90	0.65	1.23	1.01	1.48	1.29	1.33
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.15$	0.56	0.57	0.86	0.78	0.58	1.24	0.98	1.58	1.08	1.36
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.18$	0.62	0.56	0.91	0.90	0.65	1.27	1.06	1.56	1.25	1.39
$Beta(1.1,1.1)$	$\mu=0.29, SD= 0.2$	0.57	0.99	0.75	0.85	0.57	1.26	1.27	1.53	1.24	1.37
$\mu=0.67, SD=0.1$	$Beta(1.1,1.1)$	0.53	1.24	0.66	0.72	0.67	1.2	1.41	1.62	1.25	1.50
$\mu=0.67, SD=0.1$	$\mu=0.29, SD=0.18$	0.58	0.88	0.62	0.67	0.61	1.22	1.29	1.47	1.18	1.37
		Mean									
$Beta(1.01,1.01)$	$Beta(1.01,1.01)$	47.8	7.2	28.4	12.1	40.5	73.6	10.5	53.6	9.6	63.2
$Beta(1.1,1.1)$	$Beta(1.1,1.1)$	45.6	7.8	25.8	11.9	37.7	70.1	11.4	49.0	9.7	58.7
$Beta(1.2,1.2)$	$Beta(1.2,1.2)$	43.0	8.2	23.3	11.6	34.9	66.4	12.0	44.8	9.6	54.4
$Beta(1.3,1.3)$	$Beta(1.3,1.3)$	41.0	8.4	21.3	11.4	32.6	63.2	12.4	41.3	9.6	50.8
$Beta(1.35,1.35)$	$Beta(1.35,1.35)$	40.0	8.4	20.3	11.4	31.7	61.8	12.5	39.6	9.7	49.2
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.1$	27.3	6.8	14.7	5.8	20.5	44.2	10.0	29.4	4.8	34.1
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.15$	38.0	7.2	21.1	9.7	30.8	59.5	10.3	41.2	8.1	49.3
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.18$	27.3	6.8	14.7	5.8	20.5	64.6	8.7	48.2	7.8	56.0
$Beta(1.1,1.1)$	$\mu=0.29, SD= 0.2$	45.5	4.9	29.8	10.5	40.3	71.8	7.0	56.0	8.8	64.8
$\mu=0.67, SD=0.1$	$Beta(1.1,1.1)$	41.3	7.8	22.1	11.3	33.5	61.1	14.5	33.7	12.9	46.7
$\mu=0.67, SD=0.1$	$\mu=0.29, SD=0.18$	40.3	5.9	22.5	11.9	34.4	59.8	10.1	35.7	14.1	49.7
		Median									
$Beta(1.01,1.01)$	$Beta(1.01,1.01)$	46.6	4.2	24.9	7.6	31.5	36.4	4.2	21.7	6.2	30.1
$Beta(1.1,1.1)$	$Beta(1.1,1.1)$	44.8	4.7	22.9	8.0	32.3	35.9	6.9	18.8	6.2	29.6
$Beta(1.2,1.2)$	$Beta(1.2,1.2)$	41.8	5.5	20.3	8.5	31.2	35.3	7.3	17.1	6.2	27.6
$Beta(1.3,1.3)$	$Beta(1.3,1.3)$	39.6	5.9	17.0	8.6	28.6	34.5	7.9	15.1	6.2	26.1
$Beta(1.35,1.35)$	$Beta(1.35,1.35)$	38.8	6.1	15.8	8.6	27.5	33.8	8.5	14.2	6.5	25.4
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.1$	26.0	6.8	12.4	3.9	19.0	26.9	6.9	13.4	2.7	19.2
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.15$	37.2	6.9	16.3	6.5	30.0	32.9	7.0	17.5	5.2	25.6
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.18$	26.0	6.8	12.4	3.9	19.0	35.7	6.1	20.8	3.5	29.4
$Beta(1.1,1.1)$	$\mu=0.29, SD= 0.2$	41.6	3.1	26.3	8.0	40.3	39.4	3.7	25.4	3.9	34.3
$\mu=0.67, SD=0.1$	$Beta(1.1,1.1)$	33.3	2.7	21.5	9.4	30.2	34.4	4.5	16.0	8.0	23.5
$\mu=0.67, SD=0.1$	$\mu=0.29, SD=0.18$										
		Variance									
$Beta(1.01,1.01)$	$Beta(1.01,1.01)$	697	77	655	101	603	8988	224	8559	127	9070
$Beta(1.1,1.1)$	$Beta(1.1,1.1)$	626	52	539	83	513	8163	218	7546	110	8035
$Beta(1.2,1.2)$	$Beta(1.2,1.2)$	564	43	465	72	454	7325	212	6564	100	7029
$Beta(1.3,1.3)$	$Beta(1.3,1.3)$	514	38	407	66	409	6605	206	5714	98	6177
$Beta(1.35,1.35)$	$Beta(1.35,1.35)$	493	36	381	65	388	6275	204	5318	100	5791
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.1$	287	14	179	27	177	2959	104	1899	38	2051
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.15$	445	17	328	56	315	5446	100	4204	77	4496
$Beta(1.1,1.1)$	$\mu=0.29, SD=0.18$	287	14	179	27	177	6691	85	5663	95	6065
$Beta(1.1,1.1)$	$\mu=0.29, SD= 0.2$	663	23	491	81	525	8199	80	7309	120	7828
$\mu=0.67, SD=0.1$	$Beta(1.1,1.1)$	486	94	214	66	498	5869	417	2983	261	4883
$\mu=0.67, SD=0.1$	$\mu=0.29, SD=0.18$	544	27	196	63	441	5355	170	2744	277	4609

³ Final year omitted since it is typically less precisely estimated.

Appendix B: Fits to the data for the preferred options

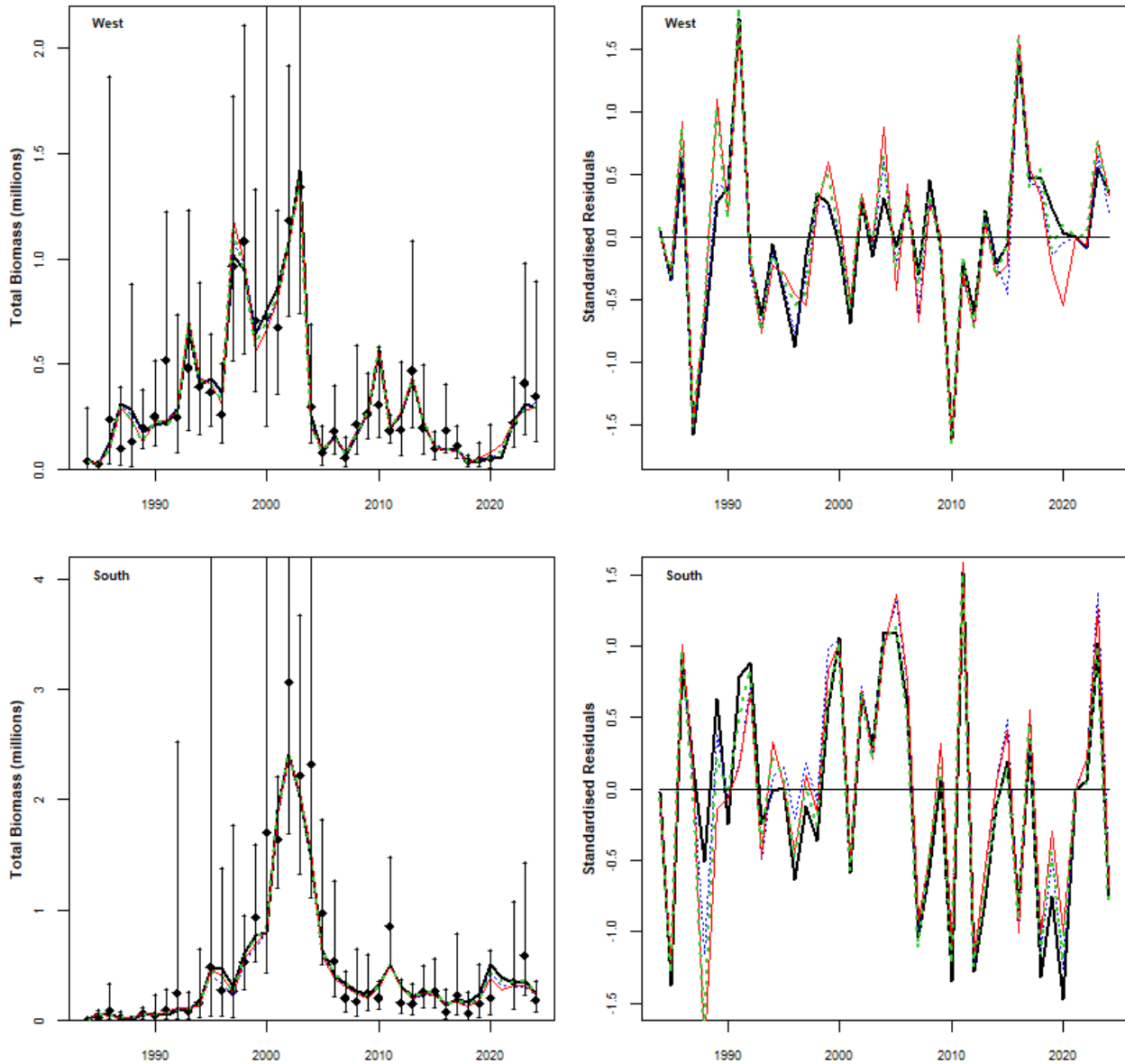


Figure B1. Acoustic survey estimated and model predicted November sardine total biomass from 1984 to 2024 for S_{noSR} of de Moor (2025b) (black), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (blue), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (red) or 0.18 (green), with $p_y \sim \text{Beta}(1.1, 1.1)$. The observed indices are shown with 95% confidence intervals. The standardised residuals (i.e. the residual divided by the corresponding standard deviation, including additional variance where appropriate) from the fits are given in the right hand plots.

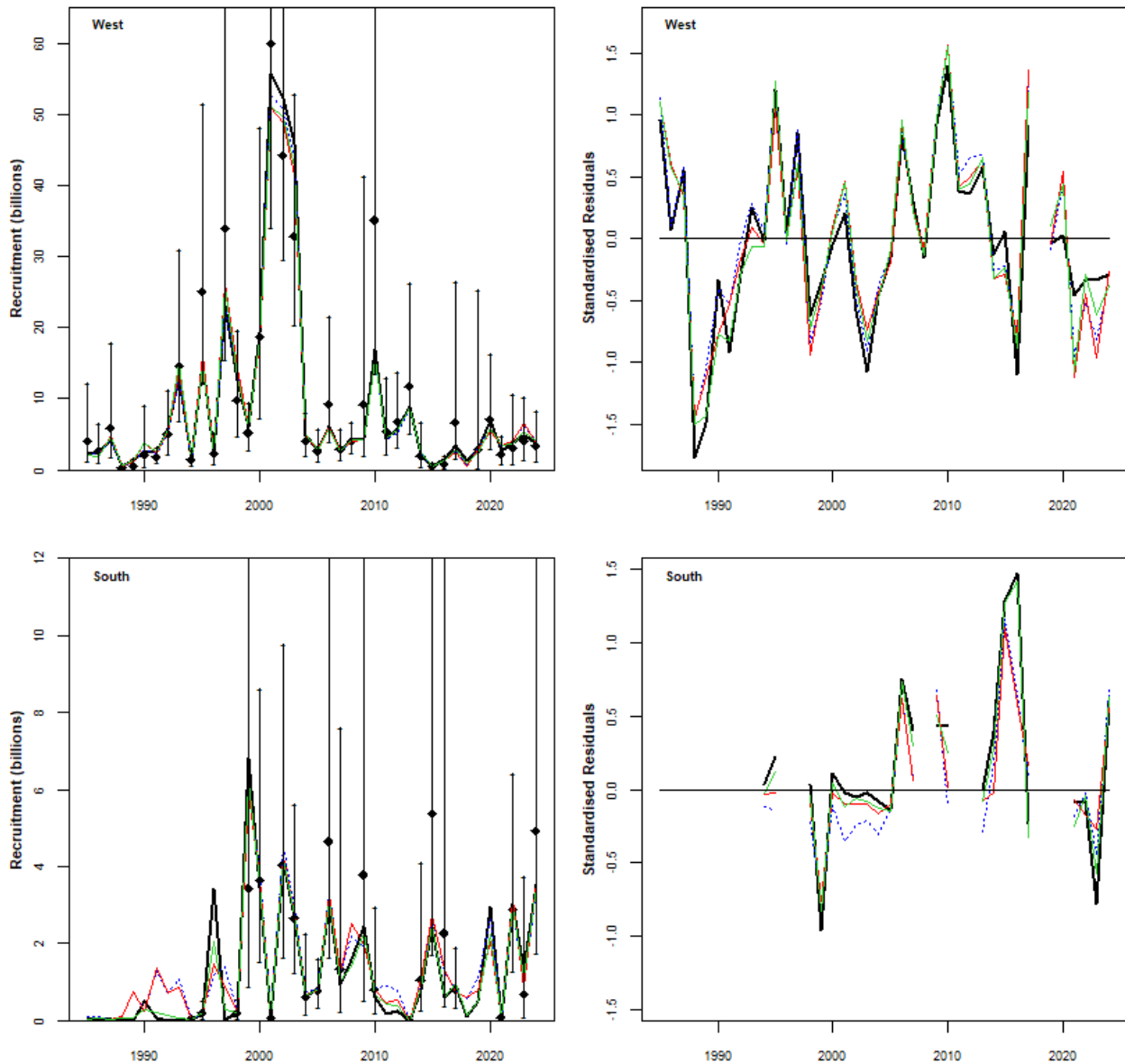


Figure B2. Acoustic survey estimated and model predicted sardine recruitment numbers from May/June 1985 to 2024 for S_{noSR} of de Moor (2025b) (black), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (blue), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (red) or 0.18 (green), with $p_y \sim \text{Beta}(1.1, 1.1)$. There was no survey observation in 2018; the model predicted value corresponds to the recruitment predicted at 8th June 2018 which is the average start date of the survey from 2016, 2017 and 2019 surveys. The survey indices are shown with 95% confidence intervals. The standardised residuals from the fit are given in the right hand plots.

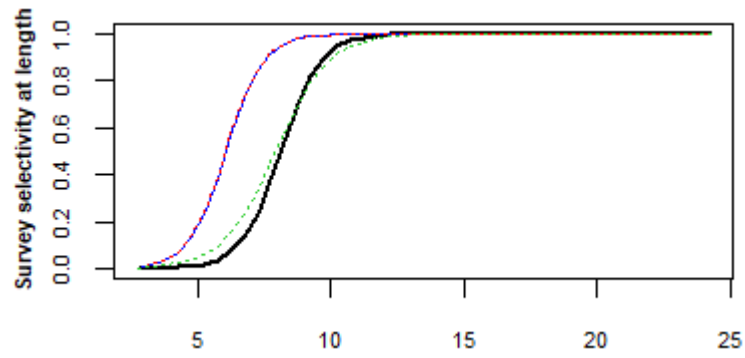


Figure B3. The model estimated November survey selectivity at length for S_{noSR} of de Moor (2025b) (black), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (blue), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (red) or 0.18 (green), with $p_y \sim \text{Beta}(1.1, 1.1)$.

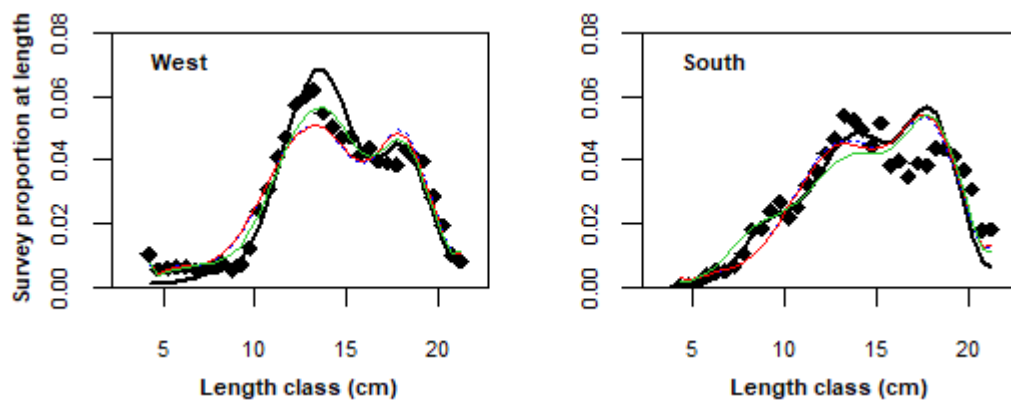


Figure B4. Average (over all years) model predicted and observed proportion-at-length in the November survey for S_{noSR} of de Moor (2025b) (black), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (blue), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (red) or 0.18 (green), with $p_y \sim \text{Beta}(1.1, 1.1)$.

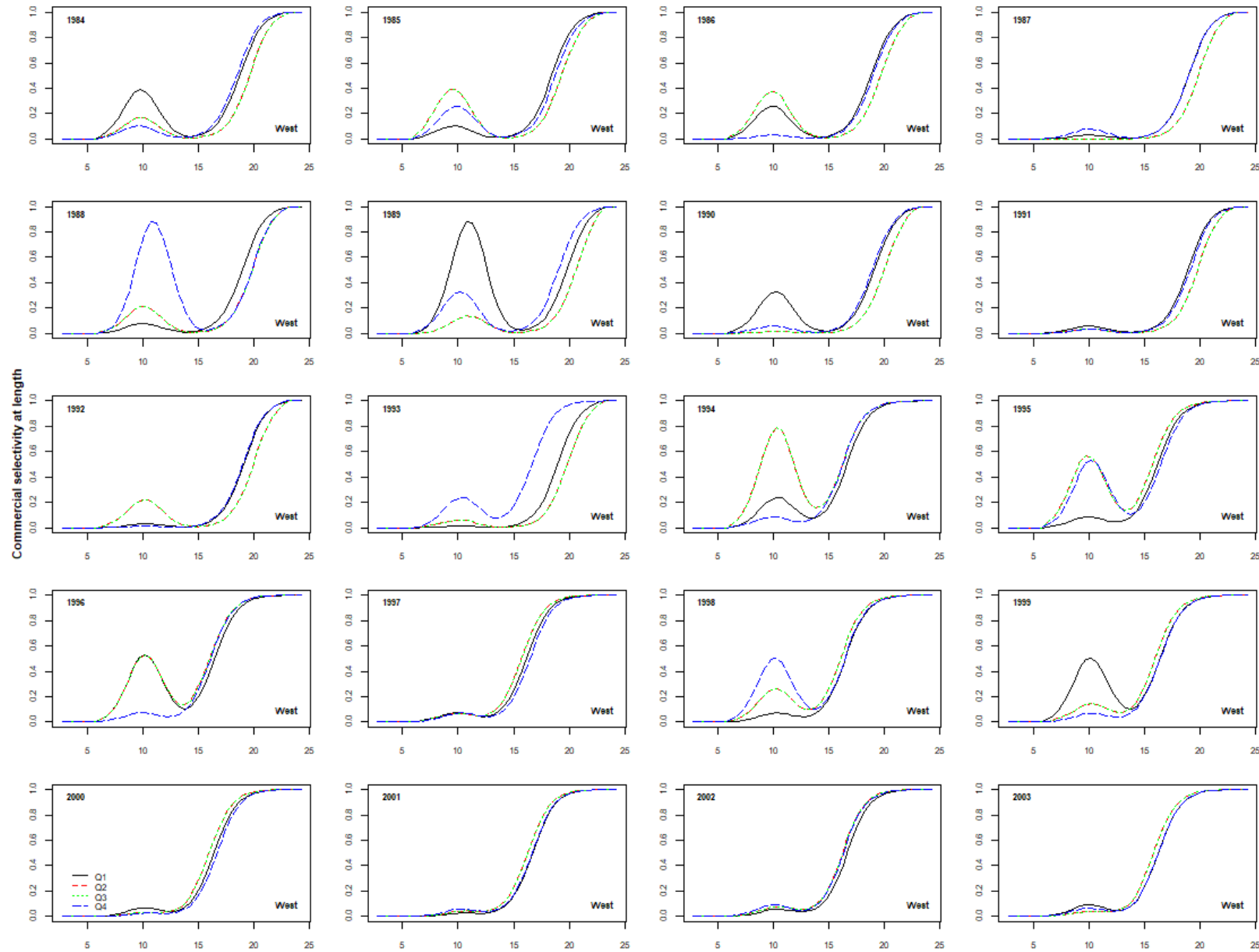


Figure B5a. The model estimated commercial selectivity at length for $p_y, q_y \sim \text{Beta}(1.35, 1.35)$.

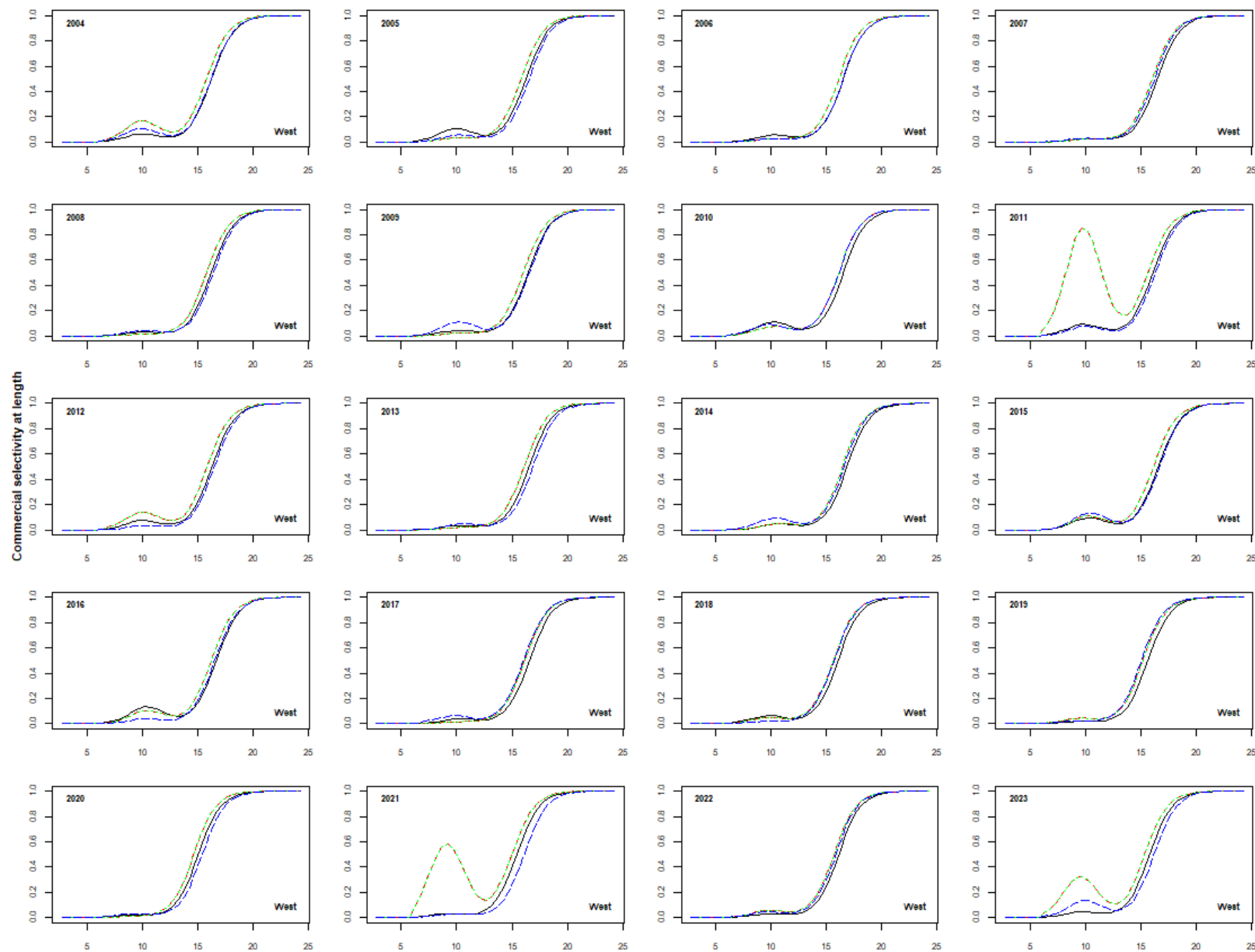


Figure B5a (continued).

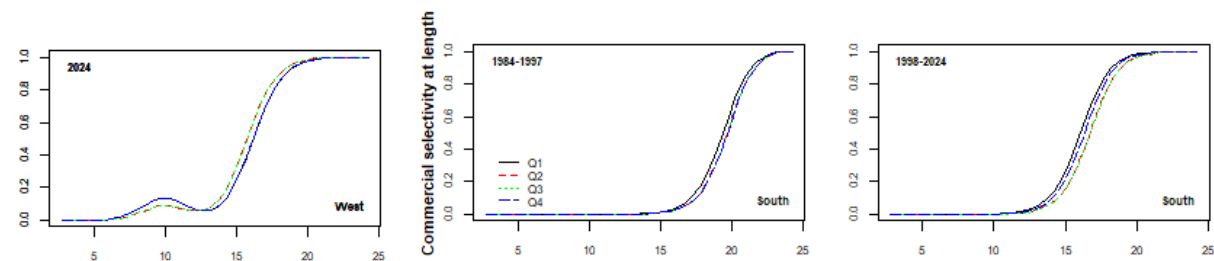


Figure B5a (continued).

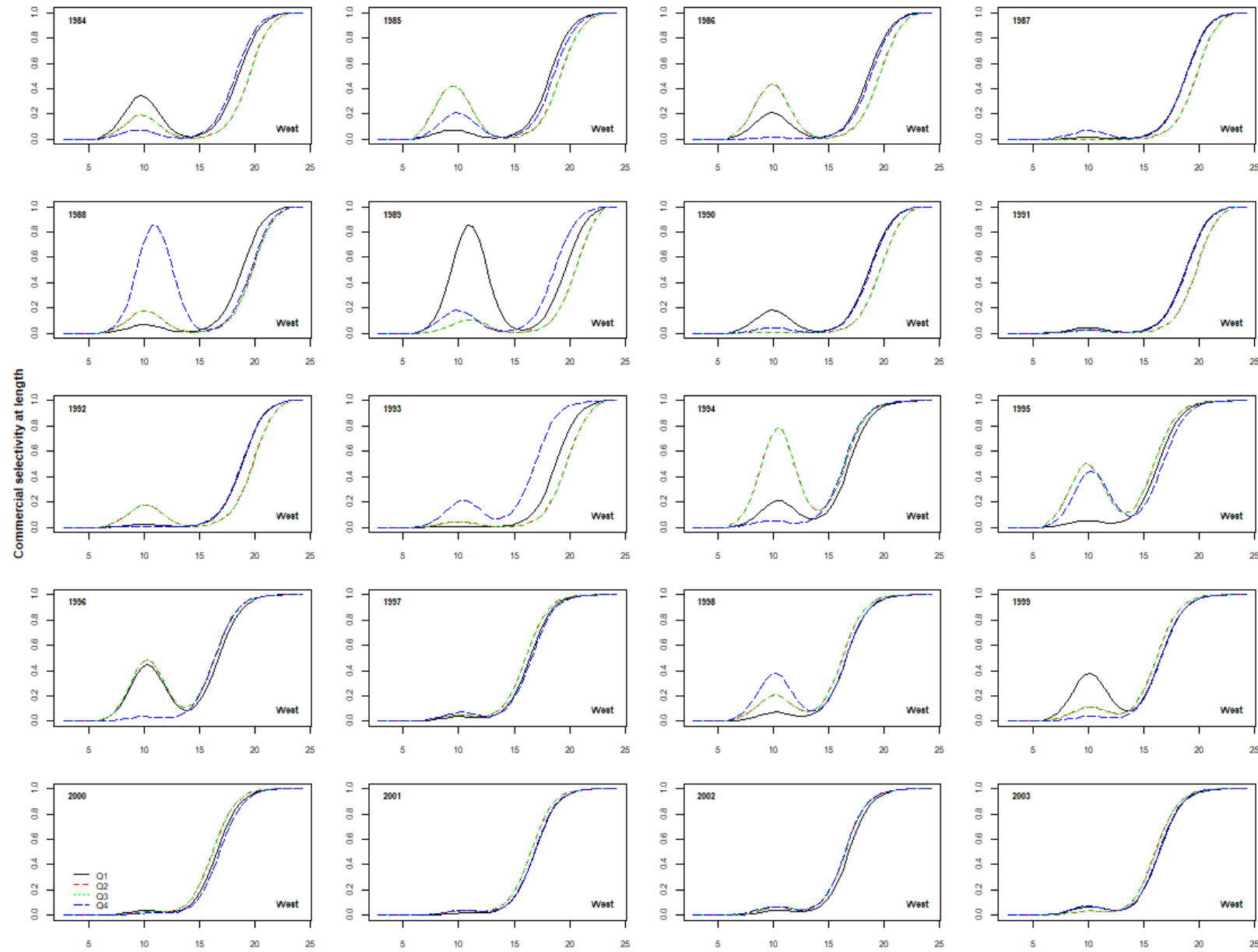


Figure B5b. The model estimated commercial selectivity at length for a Beta distribution for q_y with mean 0.29 and standard deviation of 0.18.

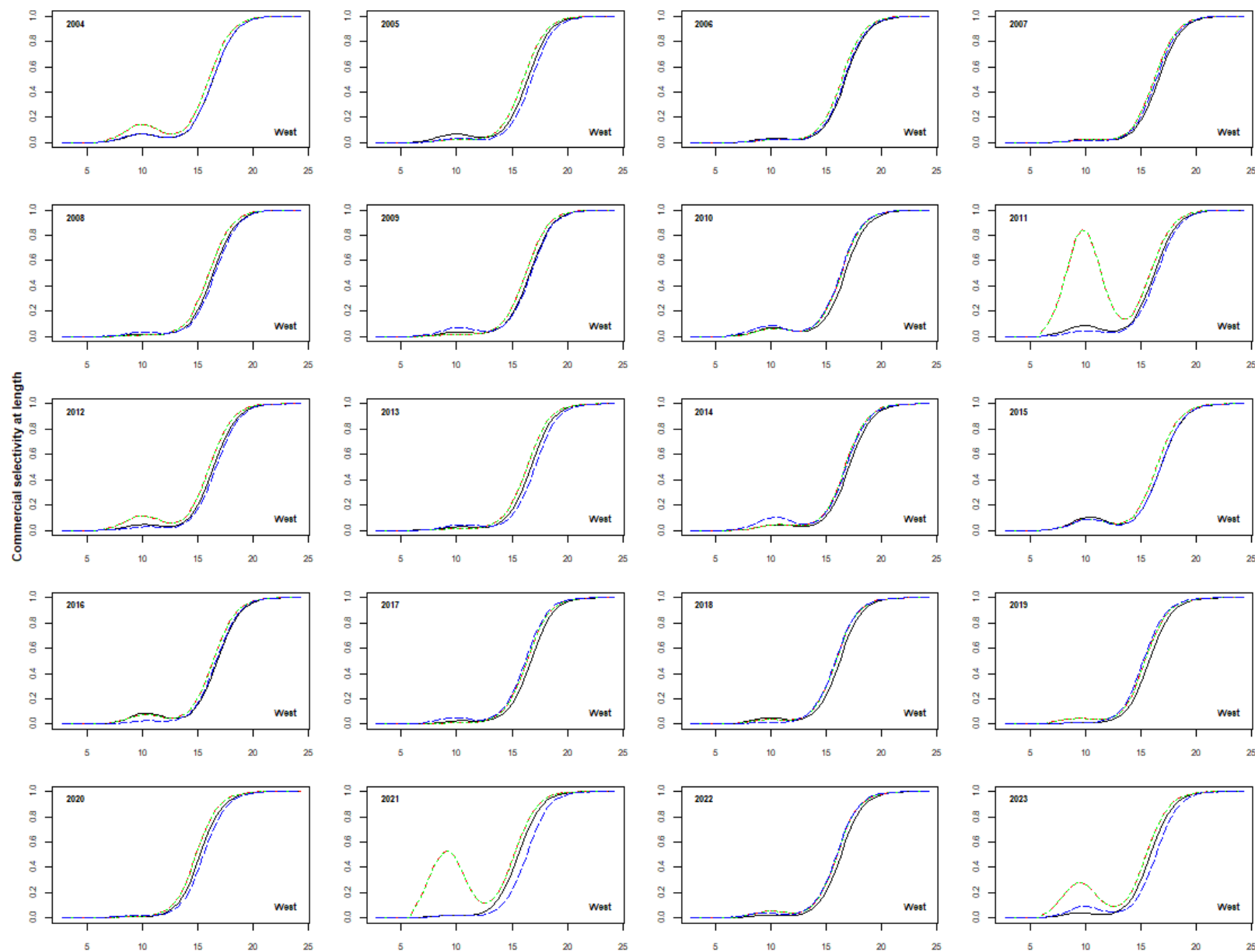


Figure B5b (continued).

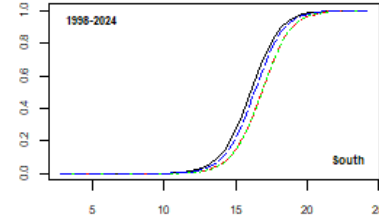
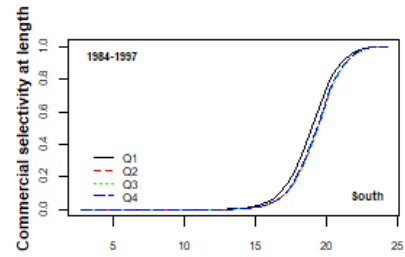
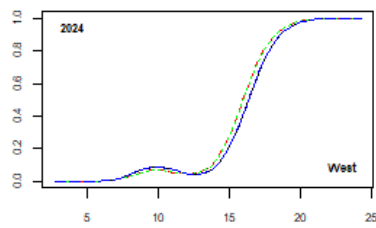


Figure B5b (continued).

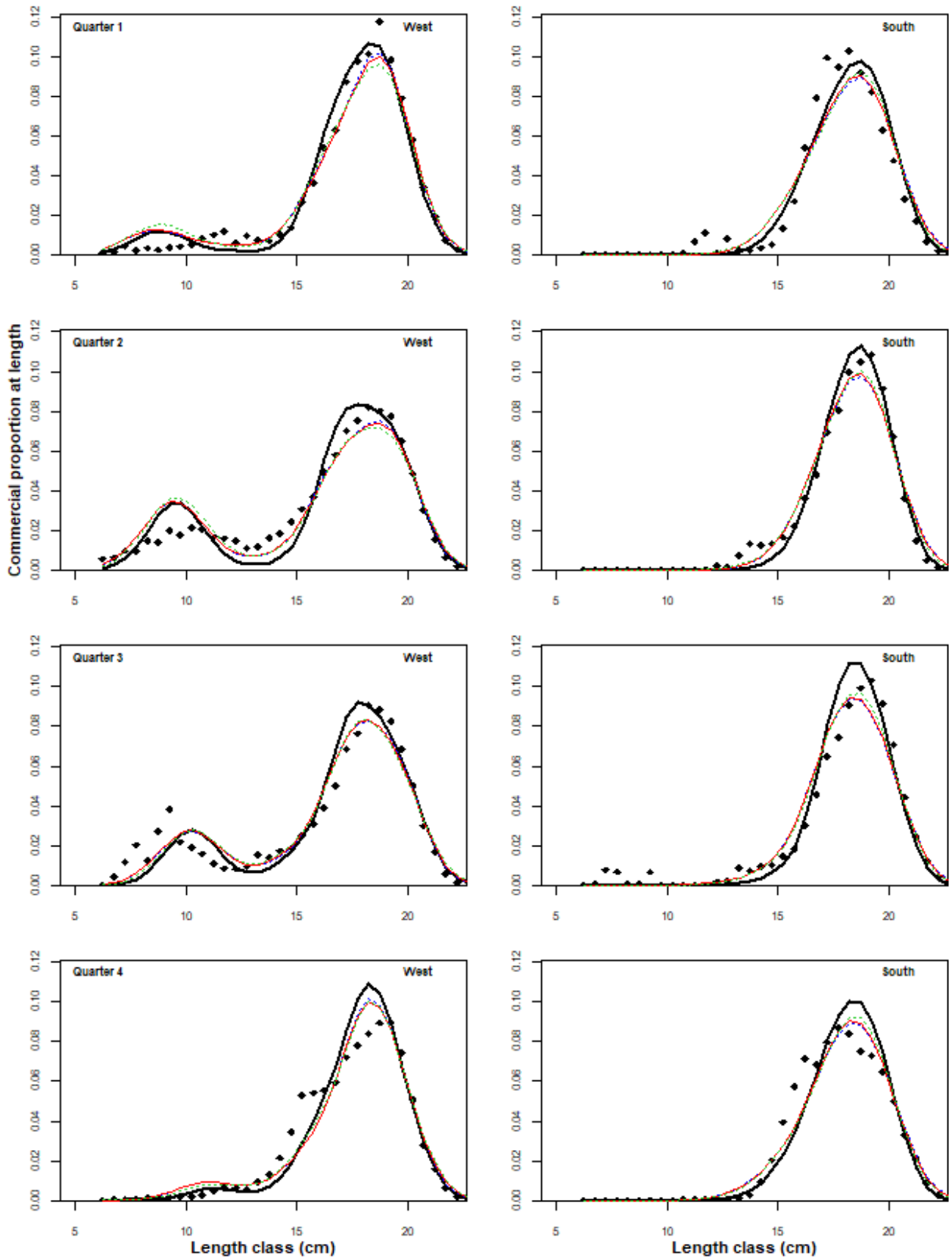


Figure B6. Average (over all years) quarterly model predicted and observed proportions-at-length in the commercial catch for S_{NoSR} of de Moor (2025b) (black), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (blue), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (red) or 0.18 (green), with $p_y \sim \text{Beta}(1.1, 1.1)$.

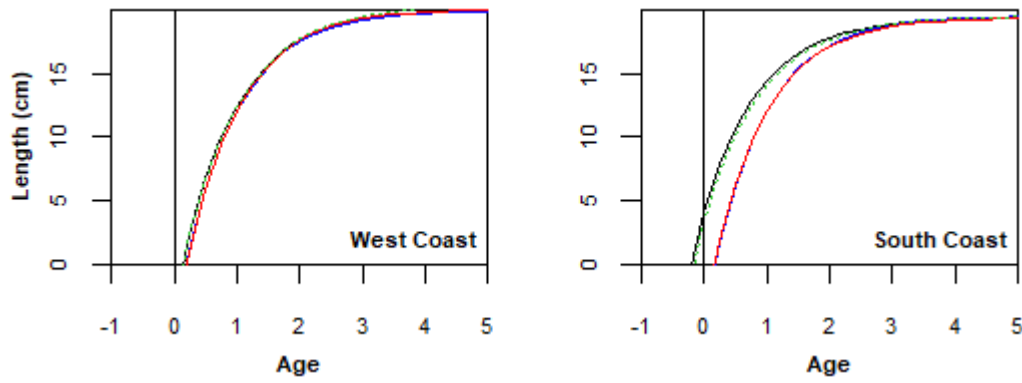


Figure B7. The von Bertalanffy growth curves for S_{noSR} of de Moor (2025b) (black), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (blue), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (red) or 0.18 (green), with $p_y \sim \text{Beta}(1.1, 1.1)$.

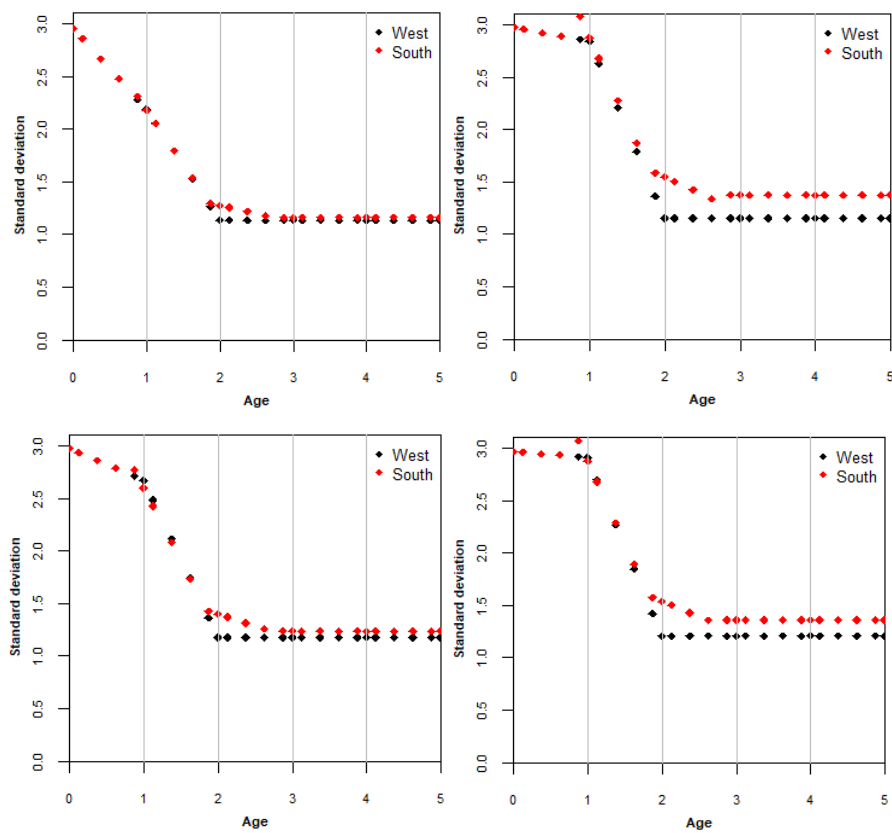


Figure B8. The estimated standard deviation about the mean length-at-age for S_{noSR} of de Moor (2025b), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$, a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 or 0.18, with $p_y \sim \text{Beta}(1.1, 1.1)$ (plotted upper left to lower right).

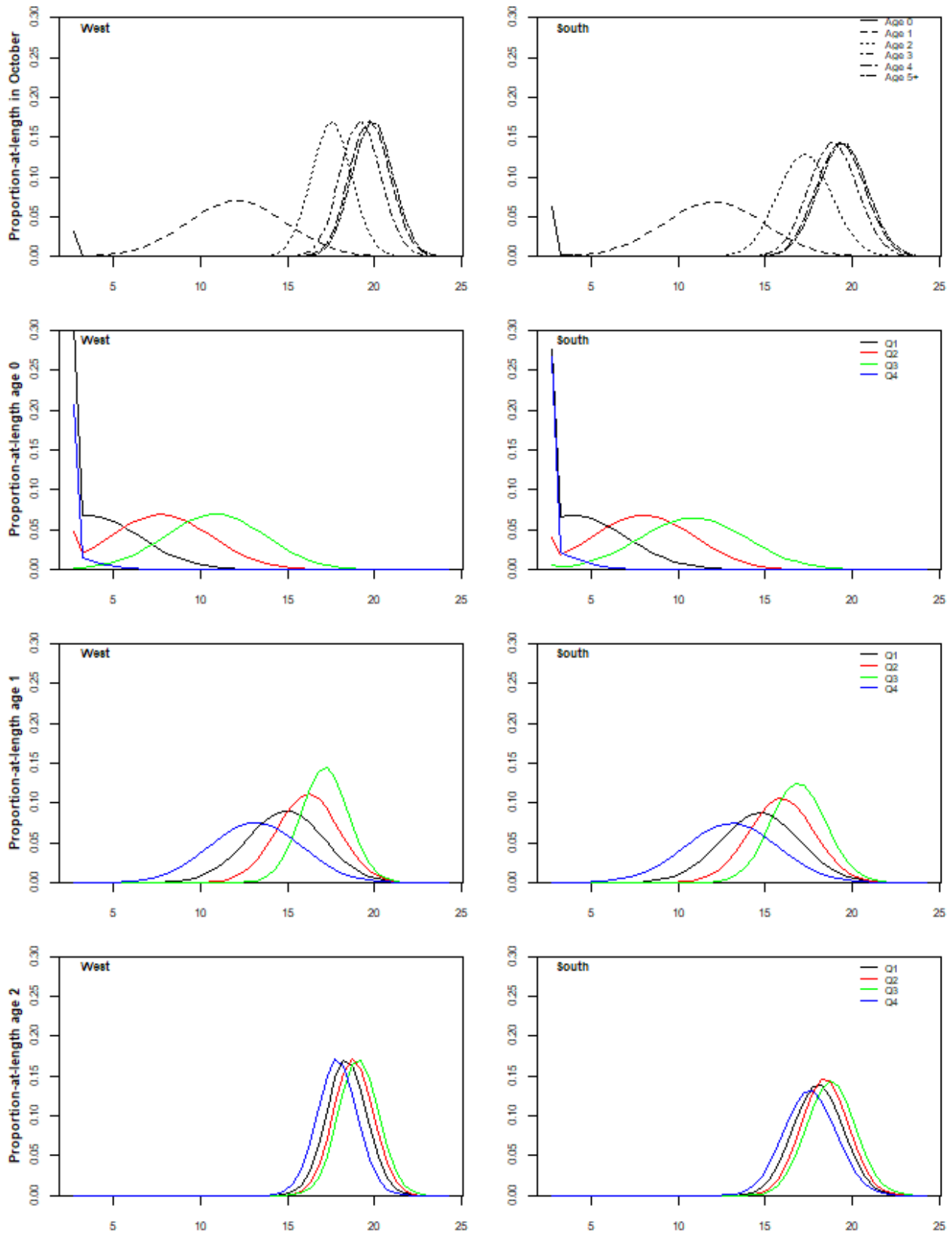


Figure B9a. The model estimated distributions of proportions-at-length for each age in 2010, given at the time of spawning (1 October, top row), and middle of each quarter of the year (corresponding to the times commercial catch is modelled to be taken for ages 0, 1 and 2) (subsequent rows). The November survey is also modelled at the middle of quarter 4. This is for $p_y, q_y \sim \text{Beta}(1.35, 1.35)$.

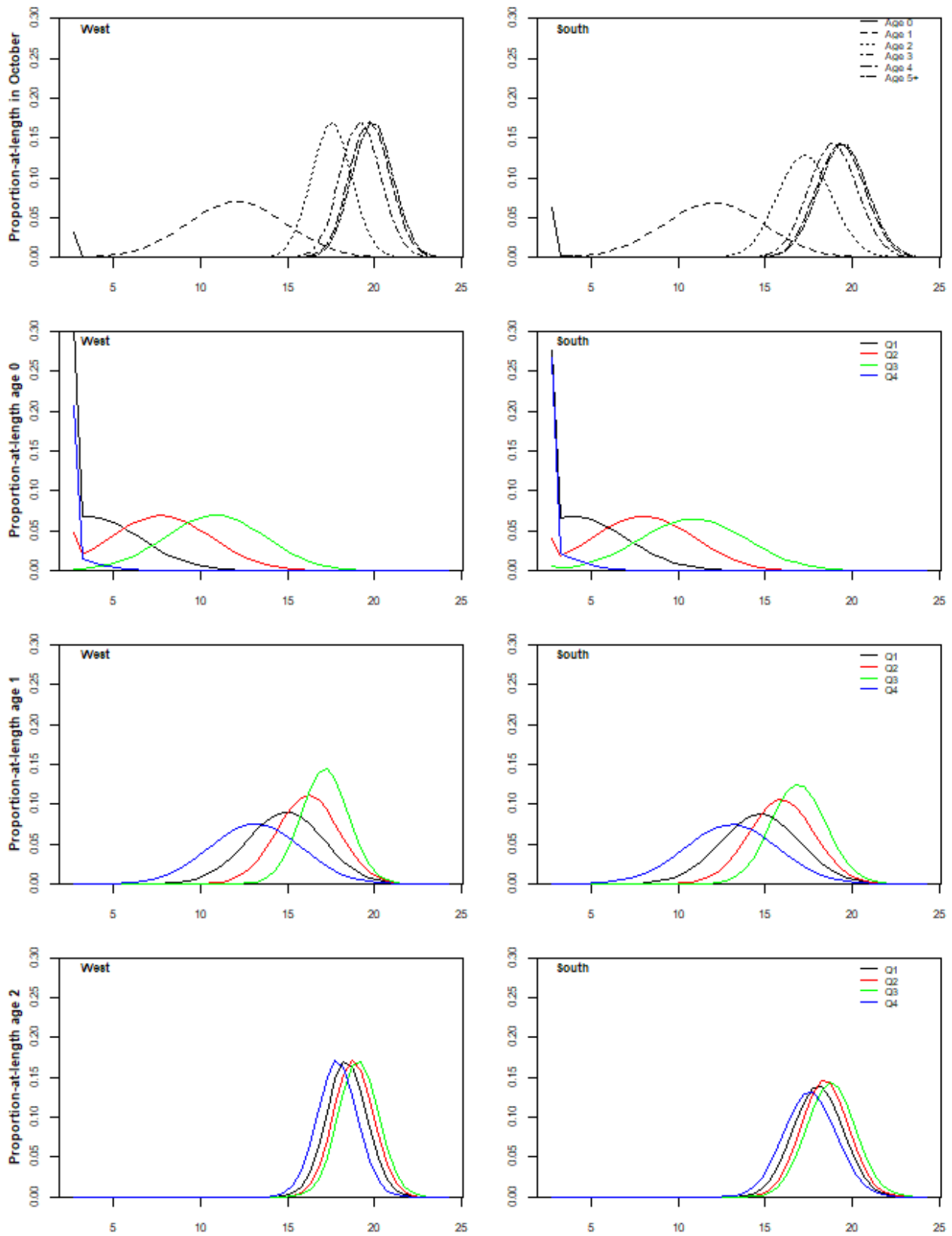


Figure B9b. A repeat of Figure 11a, but for a Beta distribution for q_y with mean 0.29 and standard deviation 0.18.

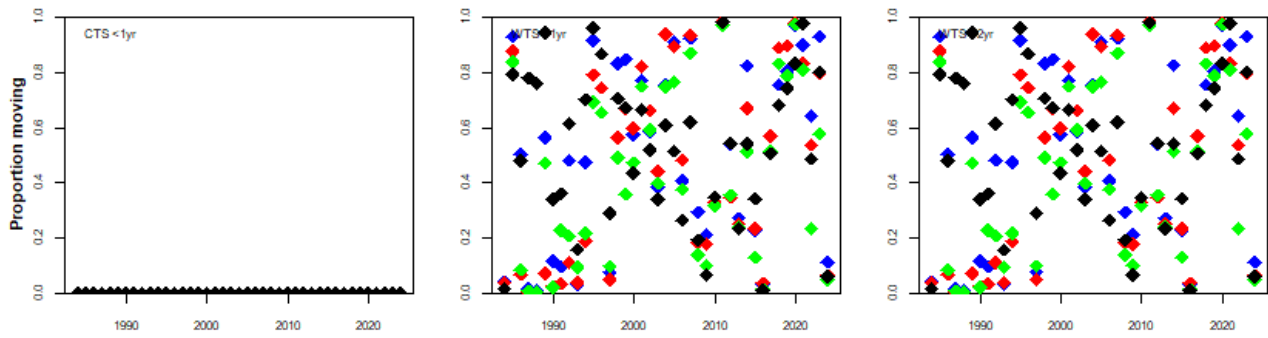


Figure B10. Model estimated proportion of <1-year-olds and <2-year-olds which move from the West Coast to the South Coast in mid-August for S_{noSR} of de Moor (2025b), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (blue), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (red) or 0.18 (green), with $p_y \sim \text{Beta}(1.1, 1.1)$.

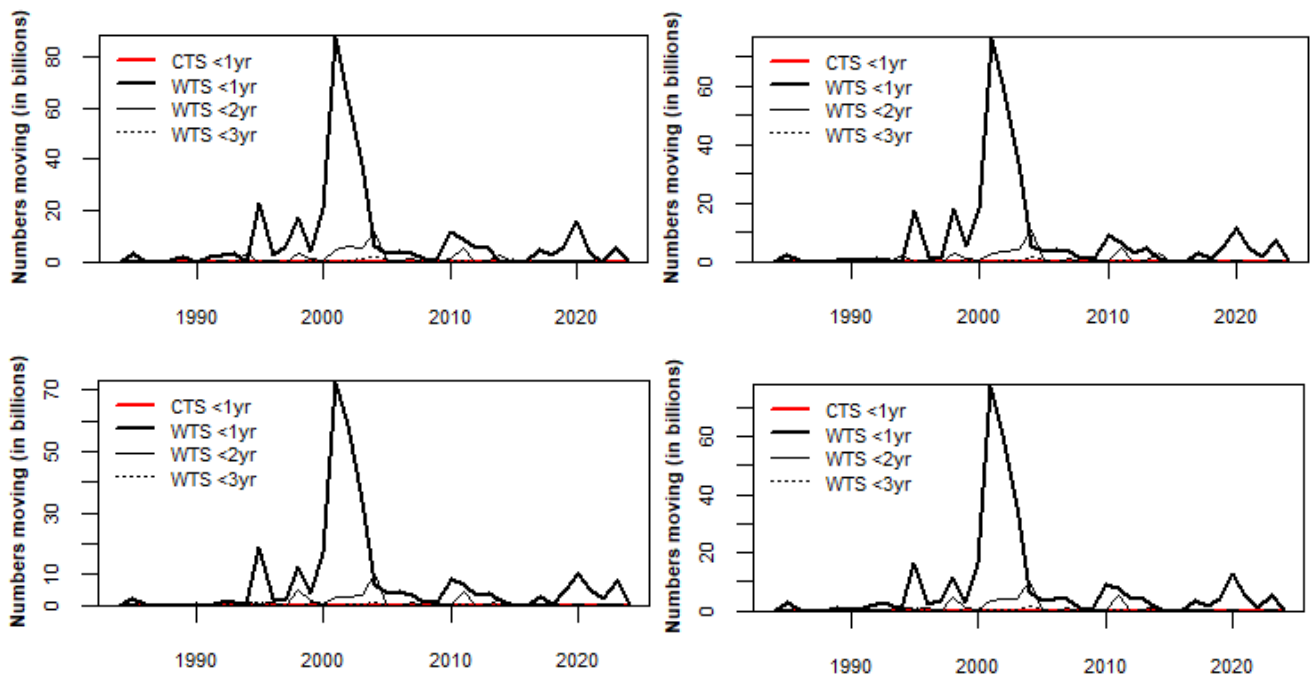


Figure B11. Model estimated number of <1-year-olds, <2-year-olds and <3-year-olds which move from the West Coast to the South Coast in mid-August for S_{noSR} of de Moor (2025b), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$, a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 or 0.18, with $p_y \sim \text{Beta}(1.1, 1.1)$ (upper to lower plots, respectively).

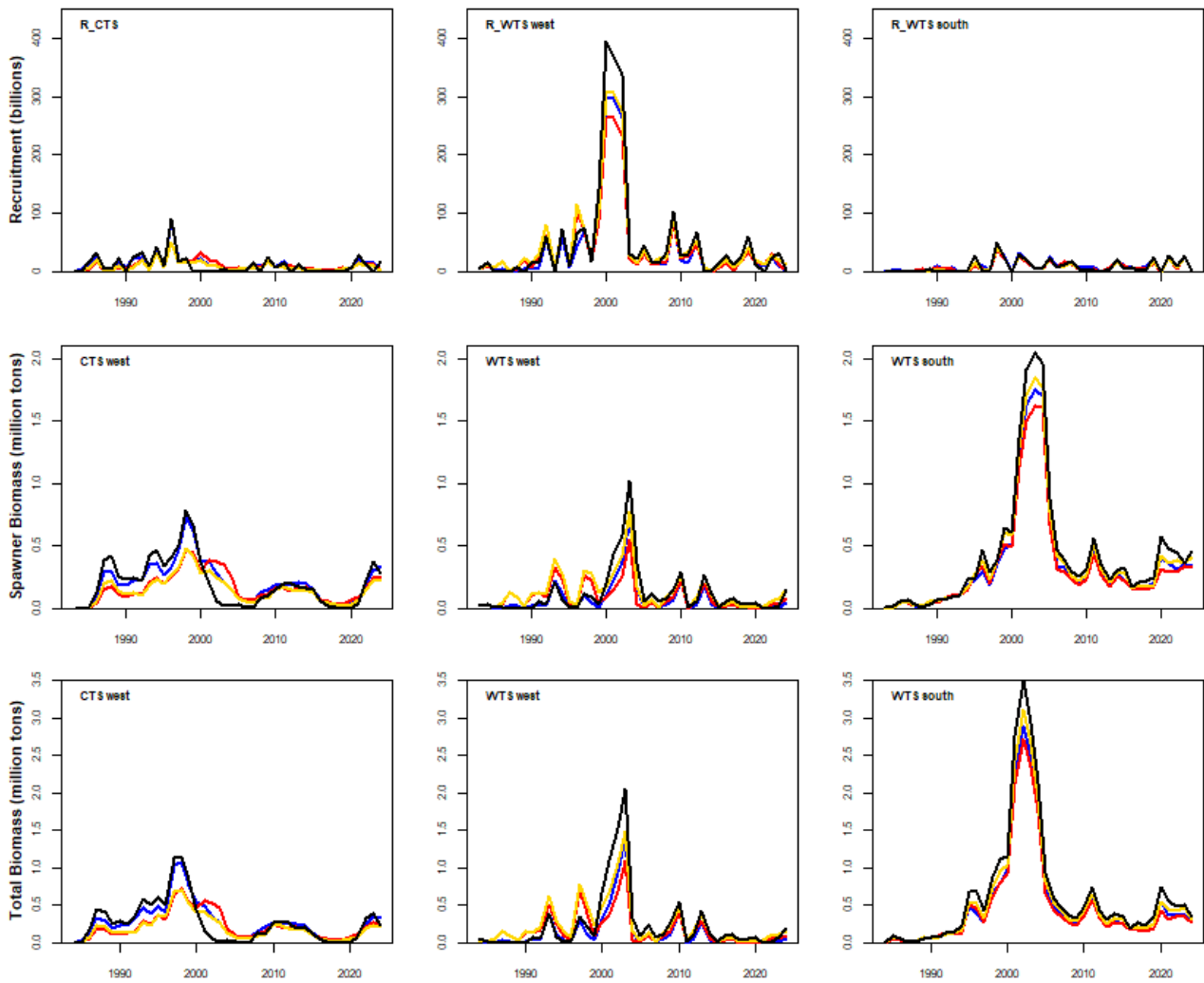


Figure B12. Model predicted sardine recruitment (top row), spawner biomass (2nd row) and total biomass (lower row) from November 1983 to November 2024 for S_{noSR} of de Moor (2025b) (black), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (blue), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (red) or 0.18 (gold), with $p_y \sim \text{Beta}(1.1, 1.1)$.

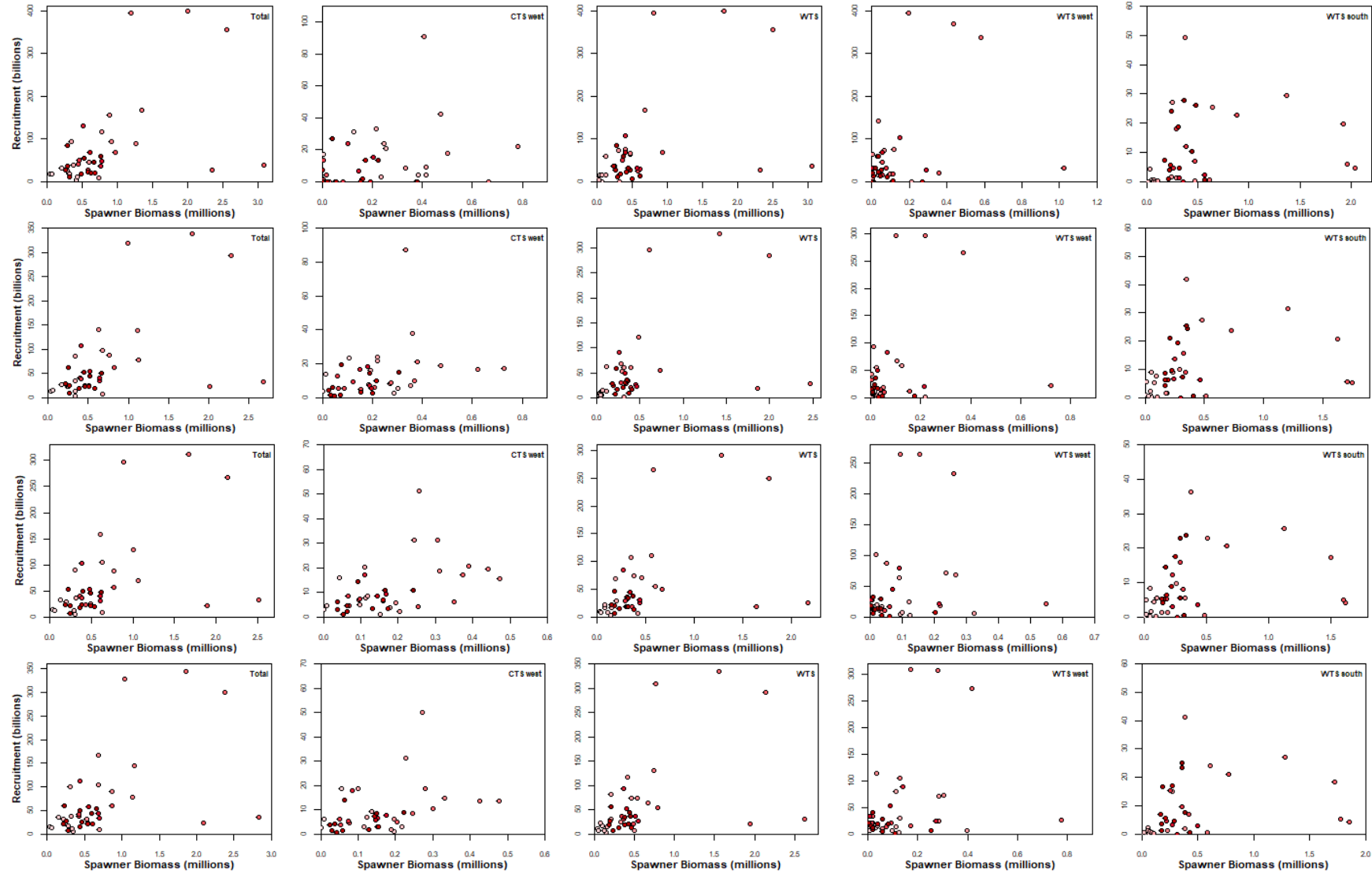


Figure B13. Model predicted sardine recruitment plotted against spawner biomass, with colouring indicating earlier (lighter) to more recent (darker) years for S_{noSR} of de Moor (2025b) (top), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (2nd row), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (3rd row) or 0.18 (lower row), with $p_y \sim \text{Beta}(1.1, 1.1)$. Stock recruitment relationships might be considered for CTS and WTS, or now for the Total recruitment instead. The right two columns show the resultant WTS recruitment by coast, which is due both to spawning by coast and passive transport of spawning products from south coast spawning to west coast recruitment.

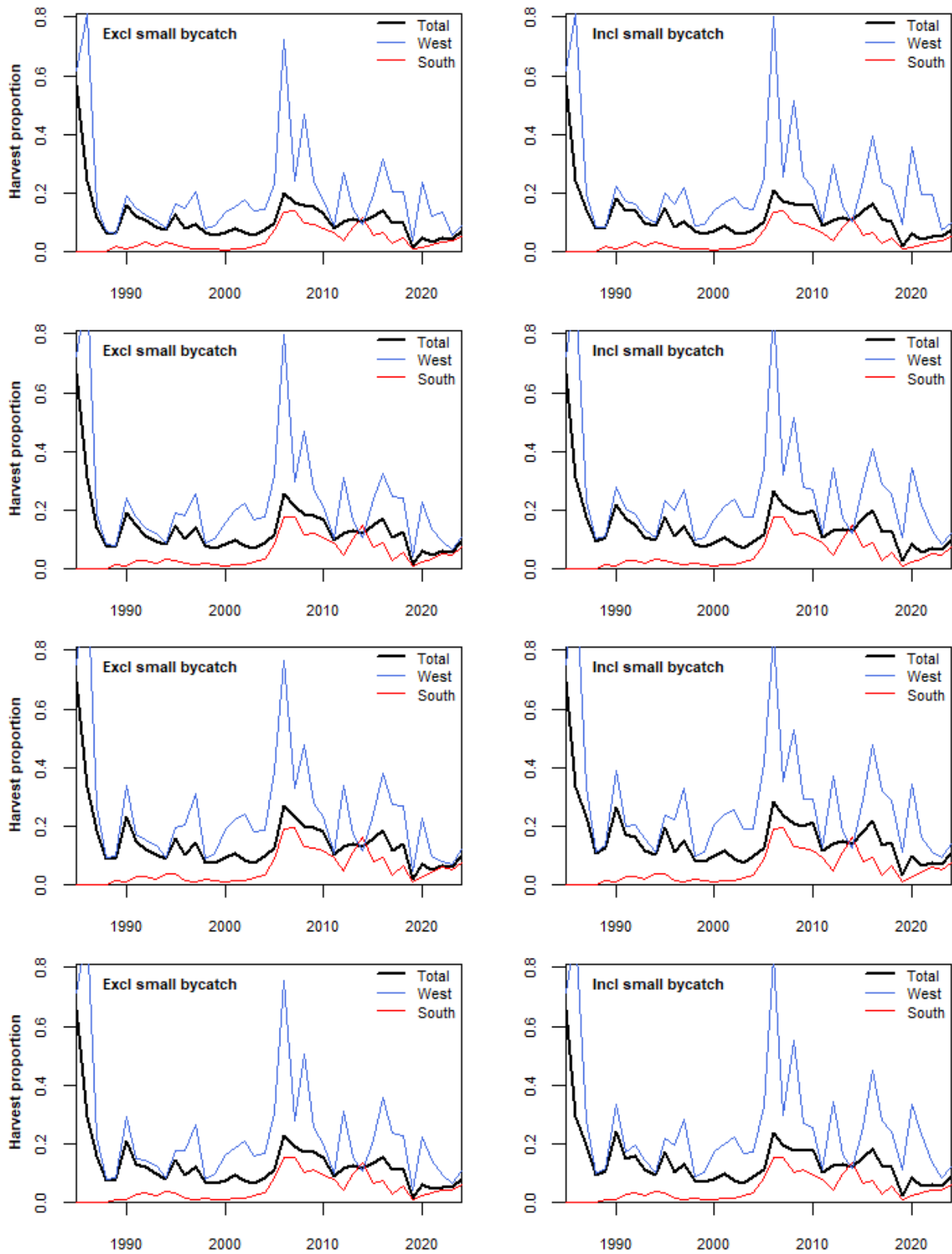


Figure B14. Historical exploitation rates by coast calculated as the annual (January – December) catch tonnage as a proportion of the November biomass of the previous year. Exploitation rates are shown excluding (left) and including (right) the small sardine bycatch. Rows show results for S_{noSR} of de Moor (2025b) (top), $p_y, q_y \sim \text{Beta}(1.35, 1.35)$ (2nd row), a Beta distribution for q_y with mean 0.29 and standard deviation 0.15 (3rd row) or 0.18 (lower row), with $p_y \sim \text{Beta}(1.1, 1.1)$.